

**K S Tharunpranav 251801359-2025-AIDS** ▾**K2****Started on** Thursday, 24 September 2026, 8:27 AM**State** Finished**Completed on** Thursday, 24 September 2026, 8:35 AM**Time taken** 7 mins 31 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int findDuplicate(int* nums, int n) {
5      int tortoise = nums[0];
6      int hare = nums[0];
7
8      do {
9          tortoise = nums[tortoise];
10         hare = nums[nums[hare]];
11     } while (tortoise != hare);
12     tortoise = nums[0];
13     while (tortoise != hare) {
14         tortoise = nums[tortoise];
15         hare = nums[hare];
16     }
17
18     return tortoise;
19 }
20
21 int main() {
22     int n;
23     if (scanf("%d", &n) != 1) {
24         return 0;
25     }
26     int* nums = (int*)malloc(n * sizeof(int));
27     if (nums == NULL) {
28         return 1;
29     }
30     for (int i = 0; i < n; i++) {
31         if (scanf("%d", &nums[i]) != 1) {
32             free(nums);
33             return 1;
34         }
35     }
36     int duplicate = findDuplicate(nums, n);
37     printf("%d\n", duplicate);
38     free(nums);
39
40     return 0;
41 }
42

```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓

	Input	Expected	Got	
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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